

$$f(x) = e^{-x/4} \cdot \tan^{-1} x$$

Del 1: Maksimalpunkt betyr $f'(x) = 0$

$$f'(x) = ((e^{-x/4})' \cdot \tan^{-1} x) + (e^{-x/4}) \cdot (\tan^{-1} x)' =$$

$$= (e^{-x/4} \cdot (-\frac{1}{4}) \cdot \tan^{-1} x) + e^{-x/4} \cdot \frac{1}{x^2+1}$$

$$f'(x) = 0.$$

$$-\frac{1}{4} e^{-x/4} \cdot \tan^{-1} x + e^{-x/4} \cdot \frac{1}{x^2+1} = 0.$$

$$e^{-x/4} \left(-\frac{1}{4} \cdot \tan^{-1} x + \frac{1}{x^2+1} \right) = 0$$

$e^{-x/4} = 0 \rightarrow$ kan aldri være 0 fordi eksp.funks er alltid positiv.

$$-\frac{1}{4} \cdot \tan^{-1} x + \frac{1}{x^2+1} = 0$$

$$\frac{1}{x^2+1} = \frac{1}{4} \cdot \tan^{-1} x \quad | \cdot 4$$

$$\frac{4}{x^2+1} = \tan^{-1} x$$

$$\tan^{-1} x - \frac{4}{x^2+1} = 0.$$

Del 2: Bestem toppunktet.

$$g(x) = \tan^{-1}(x) - \frac{4}{x^2+1} = 0$$

Vi bruker Newtons metode:

$$x_{n+1} = x_n - \frac{g(x_n)}{g'(x_n)}$$

$$g'(x) = (\tan^{-1} x)' - \left(\frac{4}{x^2+1} \right)' = \frac{1}{x^2+1} + \frac{8x}{(x^2+1)^2}$$

$x_0 = 1$! $g(1) = \arctan(1) - \frac{4}{1^2+1} = 0,7854 - 2 = -1,2146$

$$g'(1) = \frac{1}{2} + \frac{8}{4} = \frac{10}{4} = \frac{5}{2} = 2,5$$

$$\begin{aligned} * \left(\frac{4}{x^2+1} \right)' &= \frac{4' \cdot (x^2+1) - 4(x^2+1)'}{(x^2+1)^2} = \\ &= \frac{0 - 4 \cdot 2x}{(x^2+1)^2} = \frac{-8x}{(x^2+1)^2} \end{aligned}$$

$$x_1 = x_0 - \frac{g(x_0)}{g'(x_0)} = 1 - \left(\frac{-1,2146}{2,5} \right) = 1 - (-0,4984) = 1,4984$$

$$\boxed{x_1 = 1,4984}$$

$$g(1,4984) = \arctan(1,4984) - \frac{4}{(1,4984)^2 + 1} =$$

$$= 0,9793 - \frac{4}{2,2452 + 1} =$$

$$= 0,9793 - 1,2325 = -0,2532$$

$$g'(1,4984) = \frac{1}{x^2 + 1} + \frac{8x}{(x^2 + 1)^2} = \frac{1}{3,2452} + \frac{8 \cdot 1,4984}{(3,2452)^2} =$$

$$= 0,3081 + \frac{11,9872}{10,5313} = 0,3081 + 1,1382 = 1,4463$$

$$x_2 = x_1 - \frac{g(x_1)}{g'(x_1)} = 1,4984 - \frac{-0,2532}{1,4463} =$$

$$= 1,4984 + 0,175 = 1,6734$$

$$\boxed{x_2 = 1,6734}$$

$$g(x_2) = \arctan(1,6734) - \frac{4}{1,6734^2 + 1} = 1,0380 - \frac{4}{3,8002} =$$

$$= 1,0380 - 1,0525 = -0,0145$$

$$g'(x_2) = \frac{1}{x^2 + 1} + \frac{8x}{(x^2 + 1)^2} = \frac{1}{3,8002} + \frac{8 \cdot 1,6734}{(3,8002)^2} =$$

$$= \frac{1}{3,8002} + \frac{13,3872}{14,4415} = 0,2631 + 0,9269 = 1,1900$$

$$x_3 = x_2 - \frac{g(x_2)}{g'(x_2)} = 1,6734 - \left(\frac{-0,0145}{1,1900} \right) = 1,6734 + 0,0121 = 1,6855$$

$$\boxed{x_3 = 1,6855}$$

$$g(x_3) = \arctan(1,6855) - \frac{4}{1,6855^2 + 1} = 1,0390 - \frac{4}{3,8409} =$$

$$= 1,0390 - 1,0414 = -0,0024$$

$$g'(x_3) = \frac{1}{x^2 + 1} + \frac{8 \cdot 1,6855}{(x^2 + 1)^2} = \frac{1}{3,8409} + \frac{13,484}{14,752} = 0,2603 + 0,914 =$$

$$= 1,743$$

$$x_4 = x_3 - \frac{g(x_3)}{g'(x_3)} = 1,6855 - \frac{-0,0024}{1,743} = 1,6855 + 0,00137 = 1,6992$$